Design and Simulation of a Hospital Network Using Cisco Packet Tracer

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April 2026

Self-Directed Cisco Network Engineering Training Program

Abstract

This project presents the design and simulation of a complete enterprise hospital network distributed across three floors using Cisco Packet Tracer. Each department is segmented using VLAN technology to ensure security and efficient traffic management. The network integrates inter-VLAN routing, OSPF dynamic routing, DHCP services, and SSH secure access. The objective is to build a scalable, secure, and efficient network infrastructure for a multi-department hospital environment.

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1 Introduction

Modern hospital environments require robust and secure networks to support various departments and critical services. In this project, a multi-floor hospital network is designed where each department operates within its own VLAN. The network ensures secure communication, centralized management, and efficient resource utilization. Each floor contains a dedicated switch and router, interconnected using Serial DCE/DTE links and configured with OSPF dynamic routing to ensure reliable communication across all departments.

2 Network Topology

The hospital network is distributed across three floors. Each floor contains one switch for VLAN segmentation and one router for inter-VLAN routing and DHCP services. The three routers are interconnected using Serial links running OSPF dynamic routing.

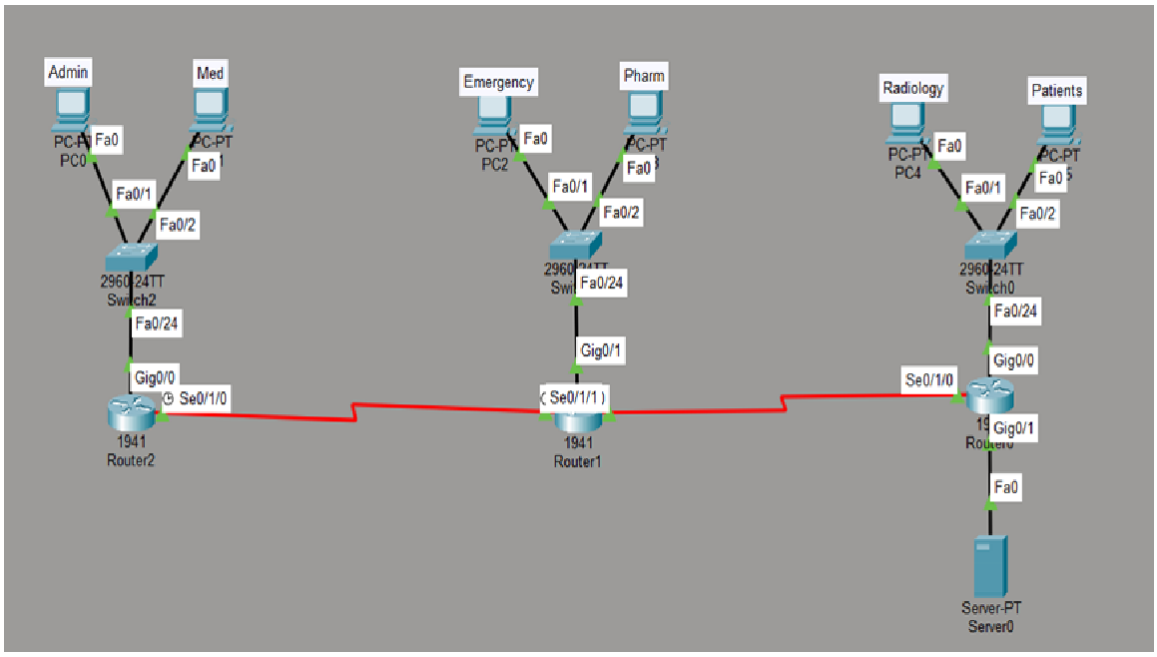

Figure 1: Hospital Network Topology — Cisco Packet Tracer

2.1 Floor 3 — Administration & Doctors

Floor 3 includes the following departments:

- Administration (VLAN 10 — 192.168.10.0/24)
- Doctors (VLAN 20 — 192.168.20.0/24)

2.2 Floor 2 — Emergency & Pharmacy

Floor 2 includes the following departments:

- Emergency (VLAN 30 — 192.168.30.0/24)
- Pharmacy (VLAN 40 — 192.168.40.0/24)

2.3 Floor 1 — Radiology & Patients

Floor 1 includes the following departments:

- Radiology (VLAN 50 — 192.168.50.0/24)
- Patients WiFi (VLAN 60 — 192.168.60.0/24)
- Server Network (192.168.100.0/24)

3 Network Design

Each floor contains one switch for VLAN segmentation and one router for inter-VLAN routing and DHCP services. The routers are connected using Serial DCE/DTE links as follows:

- R2 → R1 : 10.0.2.0/30 (R2=10.0.2.2, R1=10.0.2.1)
- R1 → R0 : 10.0.1.0/30 (R1=10.0.1.2, R0=10.0.1.1)

4 IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Description
R2	Gi0/0.10	192.168.10.1	255.255.255.0	VLAN 10 Gateway
R2	Gi0/0.20	192.168.20.1	255.255.255.0	VLAN 20 Gateway
R2	Se0/1/0	10.0.2.2	255.255.255.252	Link to R1
R1	Gi0/1.30	192.168.30.1	255.255.255.0	VLAN 30 Gateway
R1	Gi0/1.40	192.168.40.1	255.255.255.0	VLAN 40 Gateway
R1	Se0/1/0	10.0.1.2	255.255.255.252	Link to R0
R1	Se0/1/1	10.0.2.1	255.255.255.252	Link to R2
R0	Gi0/0.50	192.168.50.1	255.255.255.0	VLAN 50 Gateway
R0	Gi0/0.60	192.168.60.1	255.255.255.0	VLAN 60 Gateway
R0	Gi0/1	192.168.100.1	255.255.255.0	Server Network
R0	Se0/1/0	10.0.1.1	255.255.255.252	Link to R1
Server	Fa0	192.168.100.10	255.255.255.0	Hospital Server

Table 1: IP Addressing Table

5 Technologies Implemented

5.1 VLAN

Each department is assigned a dedicated VLAN to isolate traffic, improve security, and optimize network performance. VLANs prevent unauthorized access between departments.

5.2 Inter-VLAN Routing (Router-on-a-Stick)

Router-on-a-Stick configuration is used on each router to allow communication between VLANs within the same floor. Each VLAN is assigned a subinterface with 802.1Q encapsulation.

5.3 OSPF Routing

OSPF is configured on all three routers to ensure dynamic routing and connectivity between all floors. All networks are announced in Area 0.

5.4 DHCP

Each router is configured as a DHCP server to automatically assign IP addresses to devices in its VLANs. Addresses .1 through .9 are excluded to reserve them for network equipment.

5.5 SSH Configuration

SSH version 2 is configured on all three routers to allow secure remote access. Telnet is disabled. Only authenticated users with the correct credentials can access the routers remotely.

5.6 Serial Links

Serial DCE/DTE links are used to interconnect the three routers, simulating real WAN connections between hospital floors. Clock rate is configured on the DCE side.

6 Configuration Examples

6.1 VLAN Configuration (Switch2)

```
Switch(config)# vlan 10
Switch(config-vlan)# name Administration
Switch(config)# vlan 20
Switch(config-vlan)# name Medecins
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config)# interface fastethernet 0/24
Switch(config-if)# switchport mode trunk
```

6.2 Inter-VLAN Routing (R2)

```
Router(config)# interface gigabitethernet 0/0.10
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gigabitethernet 0/0.20
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

6.3 DHCP Configuration (R2)

```
Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
Router(config)# ip dhcp pool Administration
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
```

6.4 OSPF Configuration

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# network 10.0.2.0 0.0.0.3 area 0
```

6.5 Serial Link Configuration

```
Router(config)# interface serial 0/1/0
Router(config-if)# ip address 10.0.1.1 255.255.255.252
Router(config-if)# clock rate 64000
Router(config-if)# no shutdown
```

6.6 SSH Configuration

```
Router(config)# hostname R0-Hospital
Router(config)# ip domain-name hospital.local
Router(config)# username admin password cisco123
Router(config)# crypto key generate rsa
Router(config)# ip ssh version 2
Router(config)# line vty 0 4
Router(config-line)# login local
Router(config-line)# transport input ssh
```

7 Testing and Verification

The network was tested using several methods:

- **Same floor ping** — PC-Admin → PC-Medecins (TTL=127, 1 router traversed)
- **Different floor ping** — PC-Admin → PC-Emergency (TTL=126, 2 routers traversed)
- **Full network ping** — PC-Admin → PC-Radiology (TTL=125, 3 routers traversed)
- **DHCP verification** — All 6 PCs received IP addresses automatically
- **OSPF neighbor verification** — All routers established OSPF adjacencies
- **Server connectivity** — All departments can reach the hospital server

8 Conclusion

This project demonstrates the design and implementation of a scalable and secure enterprise hospital network. The use of VLANs, OSPF routing, DHCP, and SSH ensures efficient communication, strong security, and reliable network management across all hospital departments. The simulation in Cisco Packet Tracer validates the functionality and performance of the network. This project serves as a foundation for real-world hospital network deployments and demonstrates practical skills in enterprise network engineering.

Project completed as part of a self-directed Cisco Network Engineering training program
Percy Kamba — Ottawa, ON — April 2026